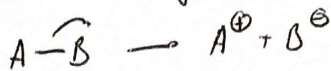


→ Rupture de liaison de façon hétérolytique



→ Rupture de liaison de façon homolytique

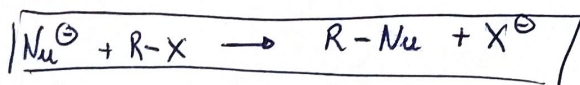
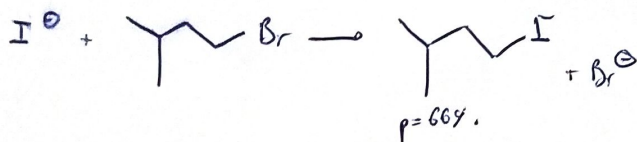


Réactions Ioniques

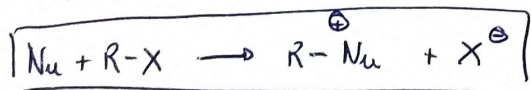
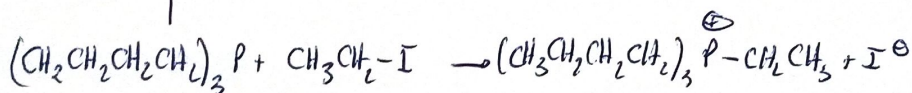
Substitution

SN

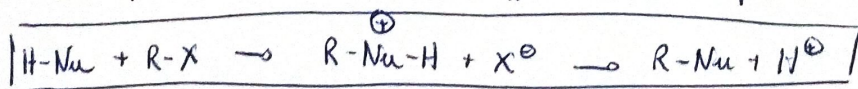
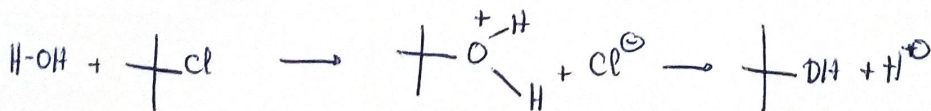
Nu chargé

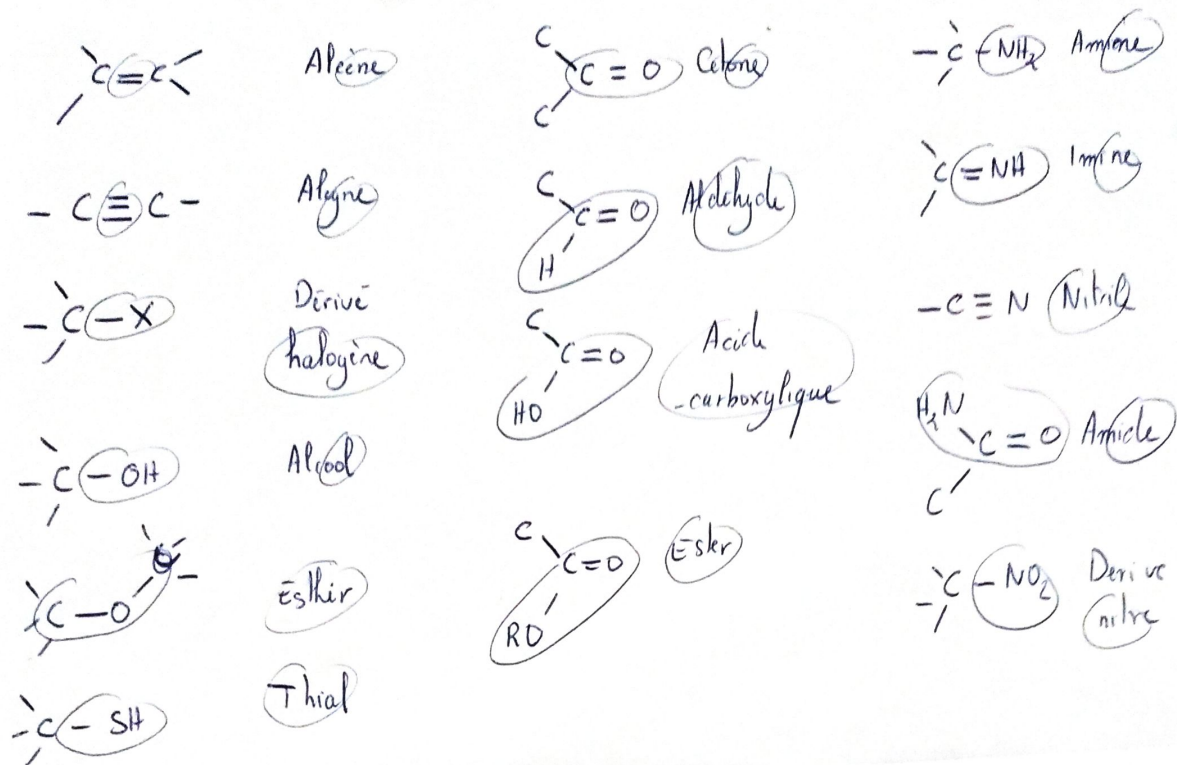


Nu non chargé



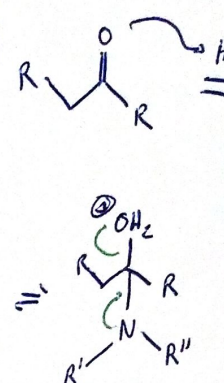
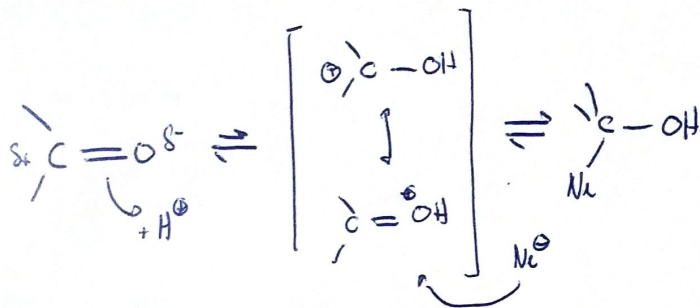
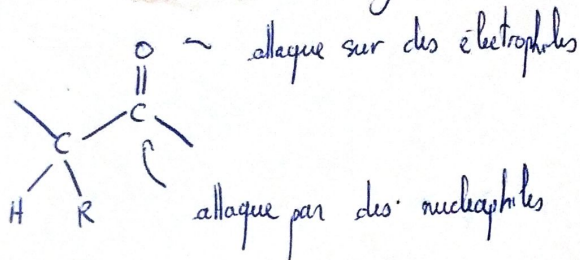
Nu H-Nu ou H-acide





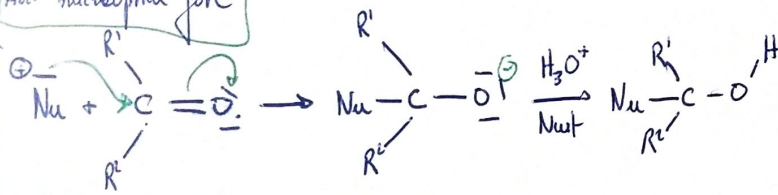
Fonction	Formule	Préfixe (suffixe)	Nom (préfixe)
Acide carboxylique	$\text{-CO}_2\text{H}$	Acide ...-oïque	-carboxy
Acide sulfonique	$\text{-SO}_3\text{H}$	Acide ...-sulfonique	sulfo
Sel	-CO_2	... carboxylate de ...	/
Anhydride carboxylique	RO-CO-OR'	Anhydride ...-oïque ...-oïque	/
Esters	-COOR	Alcenoate d'alkyle	alkoxycarboxyle
Nitrile	-CN	-nitrile	Cyano
Alcaldhyde	-CHO	-al	Formyle
Cétone	-CO-	-one	Oxo
Alcool, phénol	-OH	-al	hydroxy-
Amine	$\text{-NH}_2, \text{NHR}, \text{NR}_2$	-amine	Amino-
Dérivé halogène	-X	/	Halogéno
Dérivé nitre	-NO_2	/	Nitro

Les cétones et aldéhydes

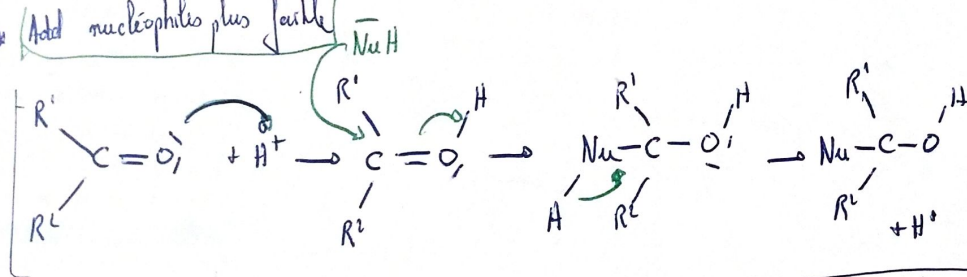


Aldolisation - cétolisation

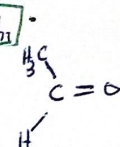
* Add nucléophile forte



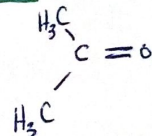
* Add nucléophile plus faible



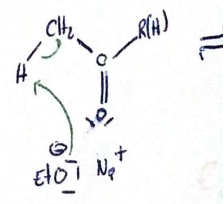
Aldolisation



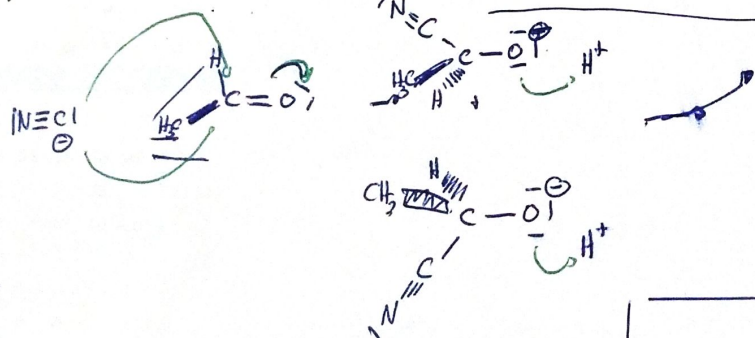
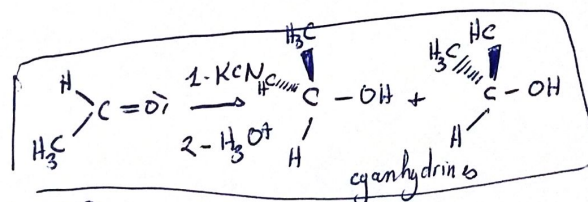
Cétolisation



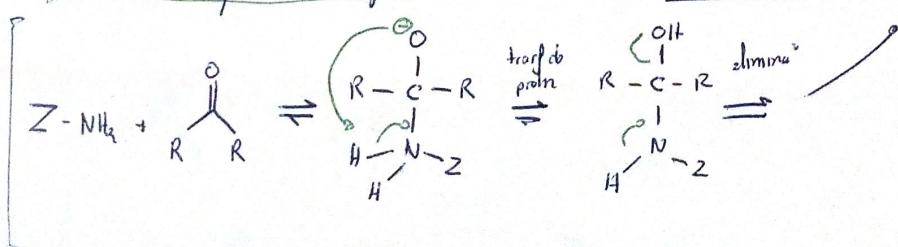
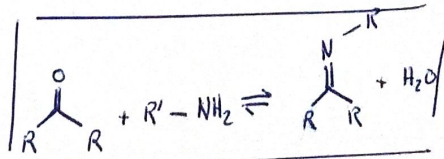
Mécanisme



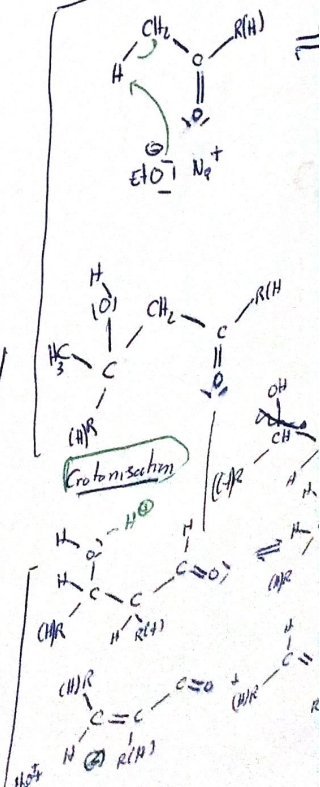
* Add ion cyanure sur carbonyle



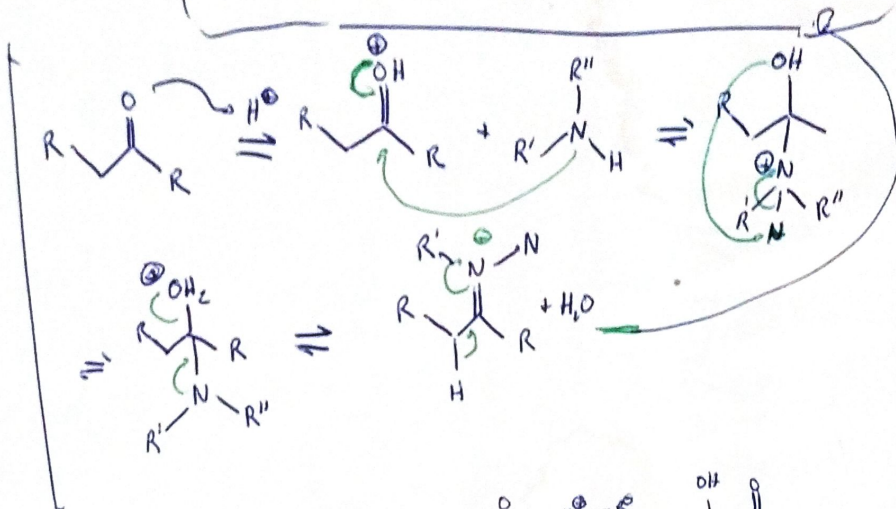
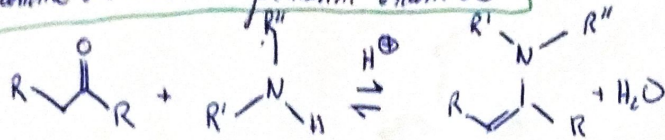
* Add amine primaire : formation d'imine



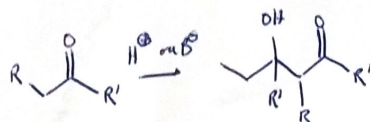
Protomécanisme



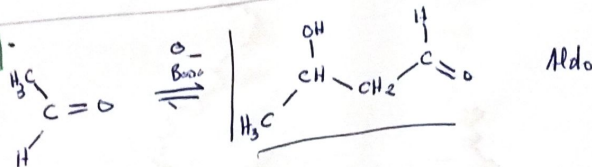
Adol amine secondaire : formation d'amine



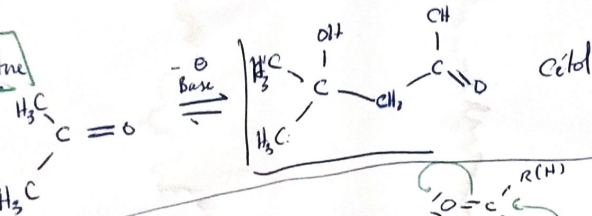
Aldolisation - cétolisation



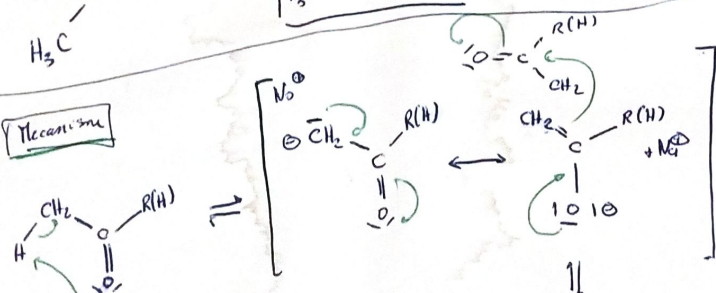
Aldolisation



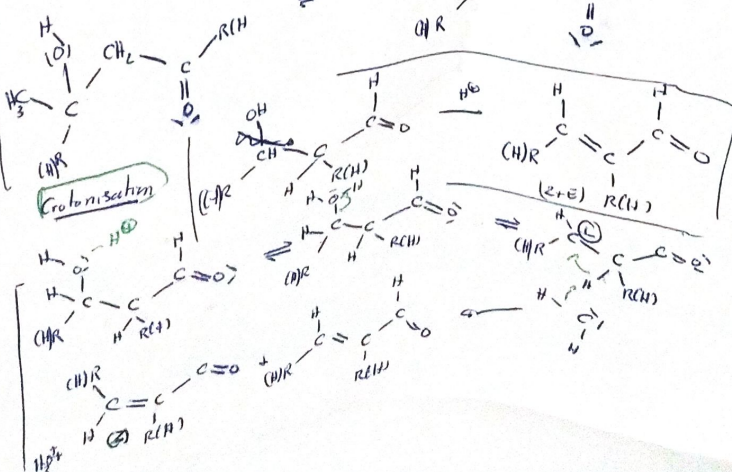
Cétolisation



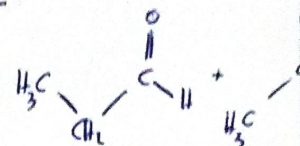
Mécanisme



Crétolisation

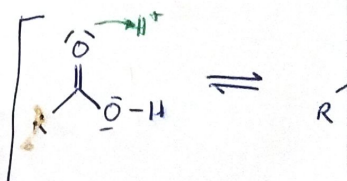


Aldolisation

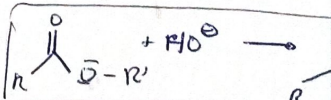


aldolhyde non énolisable

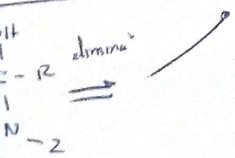
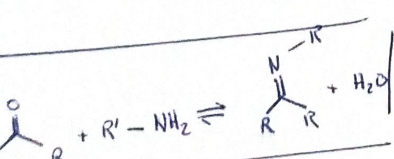
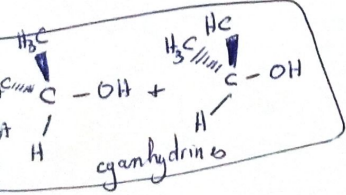
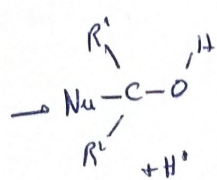
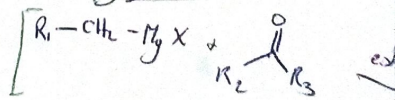
Estérification

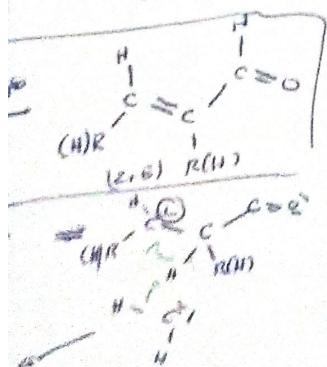


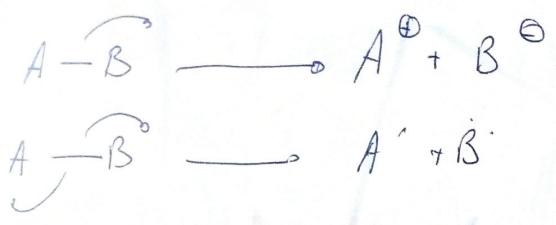
Saponification



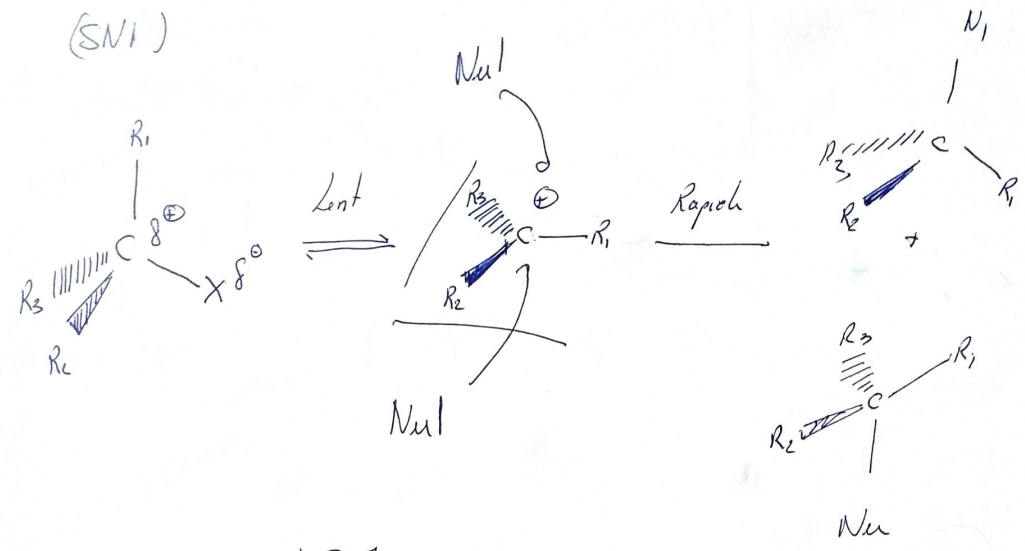
Grignard





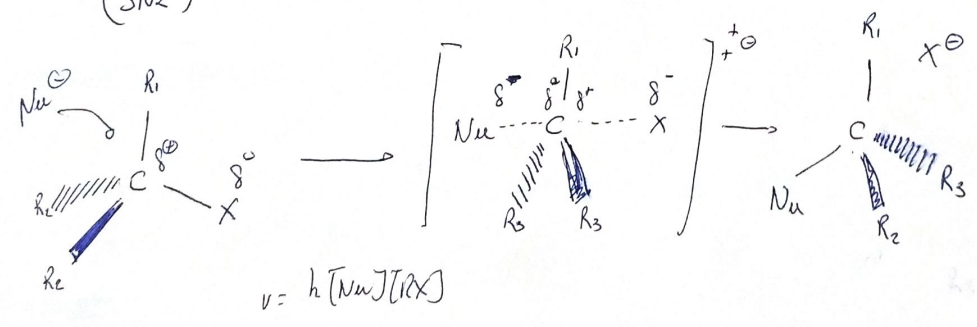


(S_N1)

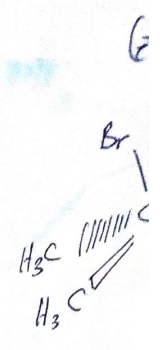


$$v = k[RX]$$

(S_N2)



$$v = k[Nu][RX]$$



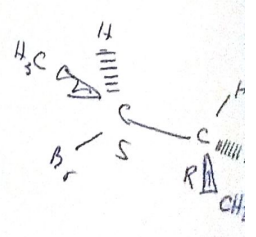
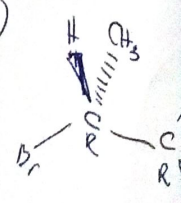
(E₁)

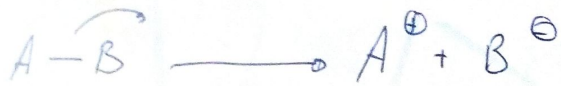


(E₁)

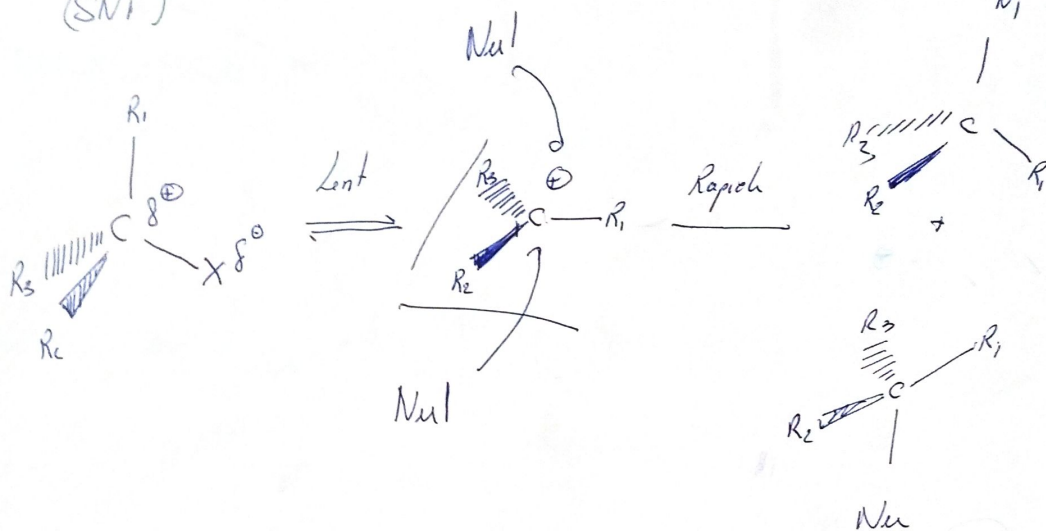


(E₂)

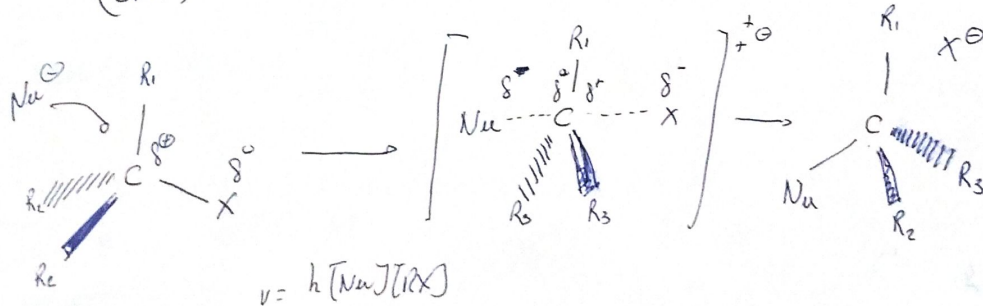




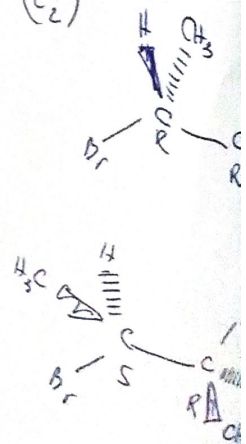
(S_N1)



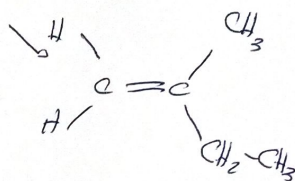
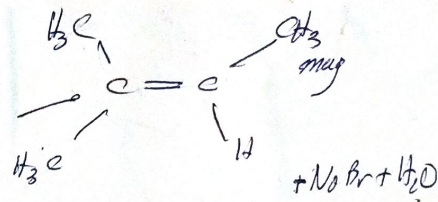
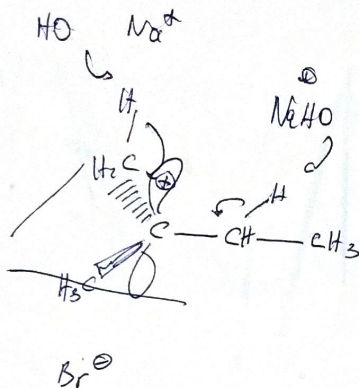
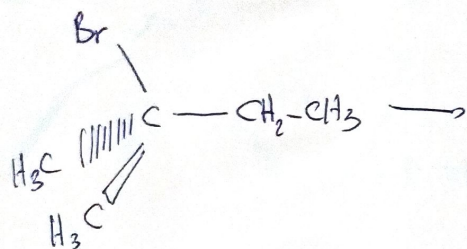
(S_N2)



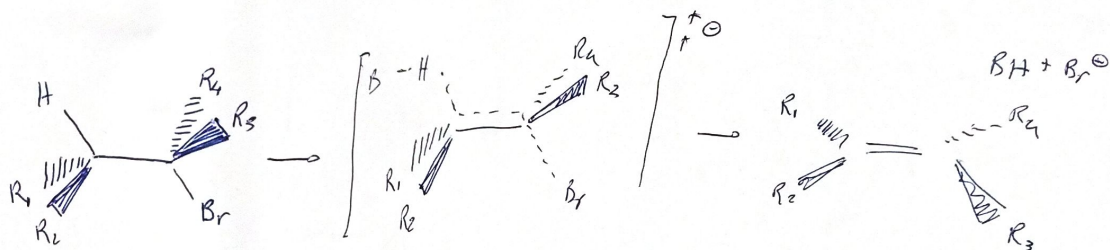
(E₂)



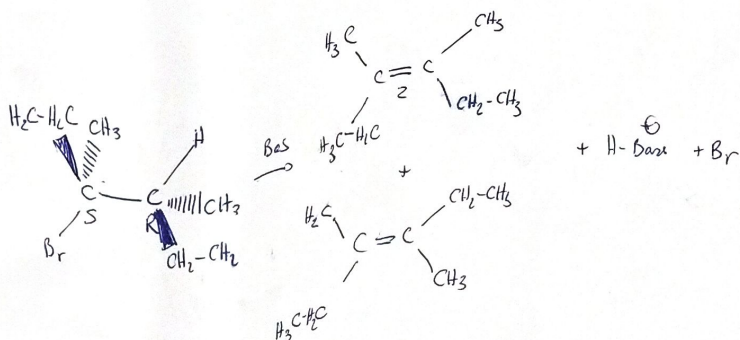
(E₁)



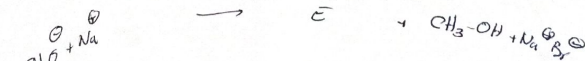
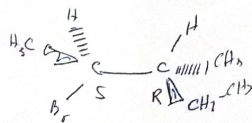
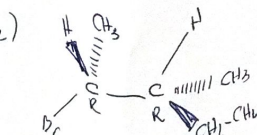
(E₂)

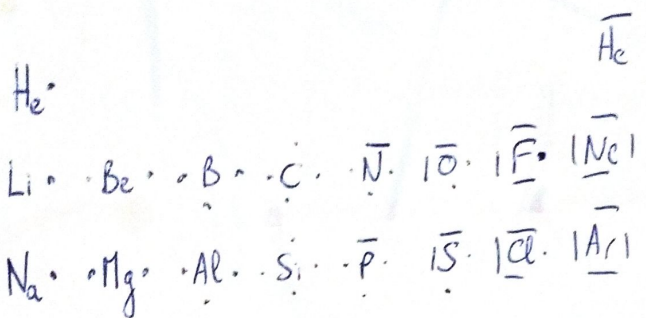


(E₁)

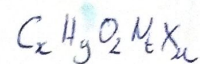


(E₂)

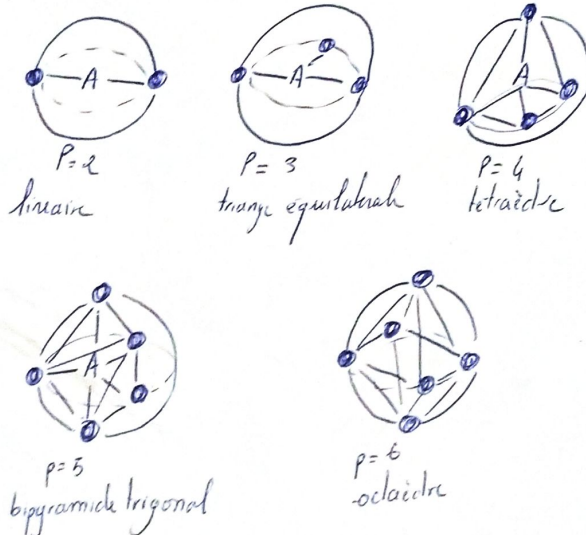




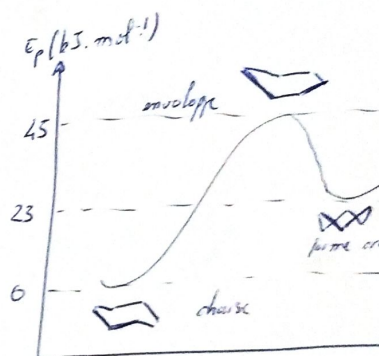
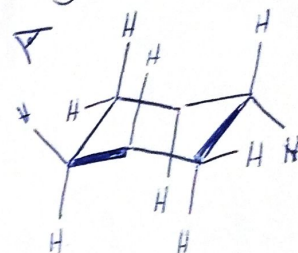
$$i = x + 1 - \frac{y}{2} + \frac{1}{2}$$



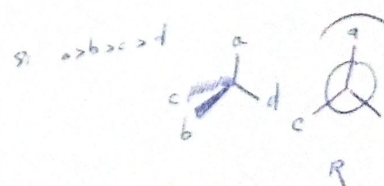
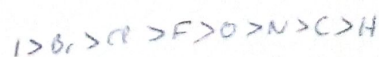
La théorie VSEPR
p: paire liant.



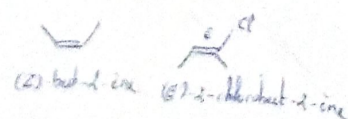
Cyclohexane



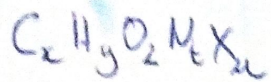
Configuration absolue R ou S



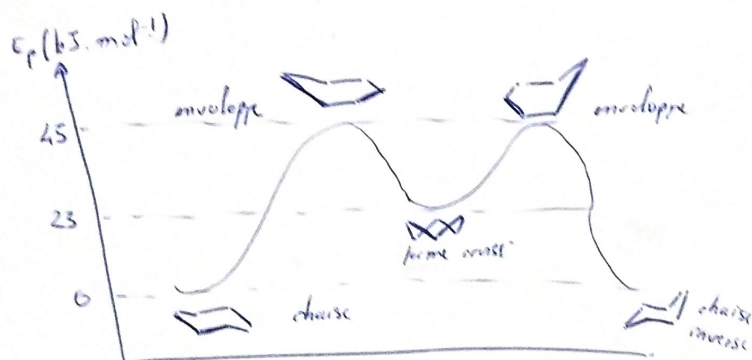
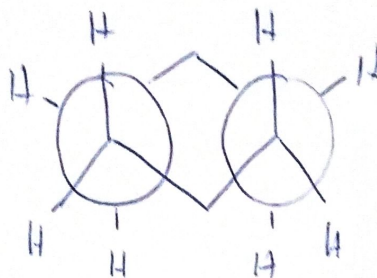
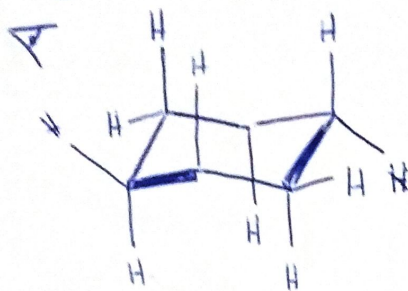
Configuration Z et E d'une double



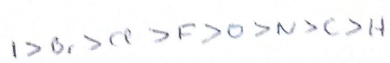
$$i = x + 1 - \frac{y}{2} + \frac{t}{2} - \frac{u}{2}$$



Cyclohexane



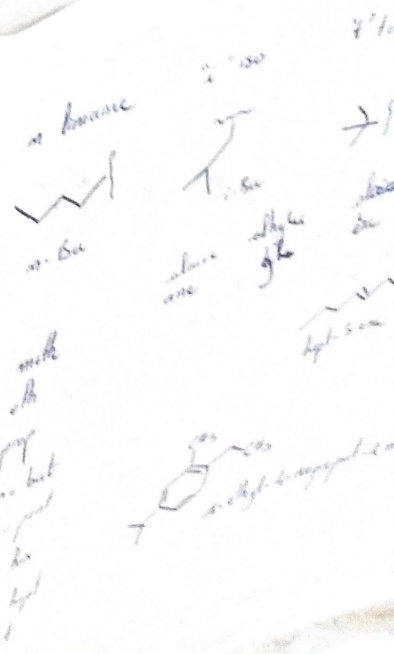
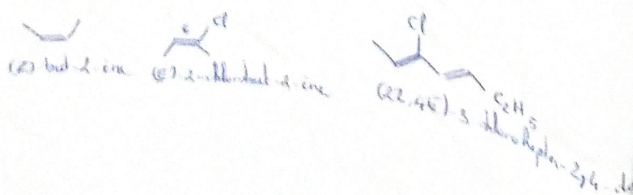
Configuration absolue R ou S d'un centre asymétrique



si: a > b > c > d



Configuration Z et E d'une double liaison



n linear



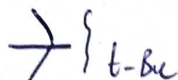
n-Bu

i' iso



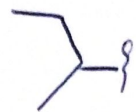
i-Bu

t' tertio



t-Bu

s' sec



s-Bu

c' cyclo

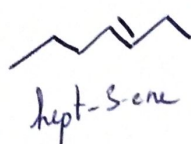


alkane

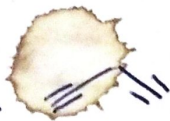
alkynes

alkene

alkyne

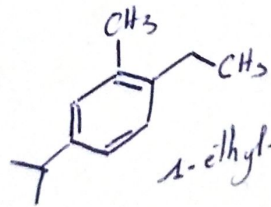


hept-5-ene



penta-1,4-diyne

- 1 methyl
- 2 eth
- 3 prop
- 4 n-but
- 5 n-pent
- 6 n-hex
- 7 n-hept
- 8 n-oct
- 9 n-non
- 10 n-undec



1-ethyl-4-isopropyl-2-methylbenzene